retrank at the NTCIR-19 R2C2 Task

Answering with Confidence as a Calibration Problem, and Why an LLM-Judged Win Still Means Something

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Abstract

We report **retrank**'s submission to the NTCIR-19 R2C2 Answering-with-Confidence (AC) subtask, scored by the Harmonic Mean of Rewards (HMR), which jointly penalises overconfidence on wrong answers (R_{O}) and underconfidence on right ones (R_{U}), alongside Accuracy and Mean Nugget Precision (MNP). **Under official scoring our four runs took the top four HMR positions among 25 runs from 7 teams (best 0.974, main system 0.963, vs. 0.761 for the strongest other run); retrank's *weakest* run (0.868) beats every other team's *best*. retrank also ranks first on Accuracy (0.969) and MNP (0.840, sweeping the top four), the MNP lead significant (α=0.05) over every team but BITEM (randomised Tukey HSD [3], B=5000).** Our thesis is that AC is a *calibration* problem disguised as a QA problem: a comparably accurate competitor (BITEM) matched our accuracy (0.92) at half our HMR (0.49), so the gap is primarily confidence discipline rather than answer quality. We locate the win in *high accuracy paired with confidence calibrated to it* (best run: mean confidence 0.956 vs. accuracy 0.953; calibration error +0.003), delivered by an answer-first pipeline whose confidence is set by an independent verifier rather than the generator. Because all R2C2 ground truth is LLM-generated (relevance judged by Qwen-3.5 and GPT-5.5) and we tuned against Claude, we confront the natural objection — "you won because you used a strong LLM" — directly: (i) our Claude self-evaluator matches the *official* (Qwen/GPT) HMR of our two main runs to within 0.0004 *in-sample*, and predicts five *held-out* competitor runs' official HMR with only weak cross-family correlation (ρ=0.60 on n=5, not significant); (ii) our pipeline run on the *cheapest* Claude tier (all-Haiku) would stay in the competitive range of the field-leading team *under self-evaluation* — so the margin is not obviously bought by model tier. We argue the lever is a largely model-agnostic calibration wrapper — it transfers across competitors' systems and, when the pipeline is swapped end-to-end (streamlined) onto three GPT models, reproduces the calibration gain (3/3 positive, one significant) — with the residual caveat that these checks use a surrogate rather than the official judge.

Keywords

answering with confidence, calibration, HMR, RAG, LLM evaluation

Team Name: retrank **Subtasks:** PR, AC

1 Introduction

The NTCIR-19 R2C2 Answering-with-Confidence (AC) subtask asks a system, for each question, to return an answer, a confidence score in $[0, 100]$, and grounded nuggets that cite passages from any team's passage-retrieval run. Systems are ranked by three official metrics — Accuracy, Mean Nugget Precision (MNP), and the Harmonic Mean of Rewards (HMR) — the last of which rewards a system for being confident *when it is right* and diffident *when it is wrong*. HMR is not a QA metric; it is a calibration metric — what the organisers term system *modesty* [5, 6]. This distinction turns out to be most of the story.

Result. Under official scoring, retrank's four runs occupy the top four HMR positions among all 25 runs from 7 teams. Our best run reaches HMR 0.974 and our main system 0.963; the strongest non-retrank run reaches 0.761. *Our weakest run (0.868) outscores every other team's best.* retrank also ranks first on Accuracy (0.969) and MNP (0.840), sweeping the top four positions on MNP as well as HMR.

Thesis. High accuracy is necessary but not sufficient in R2C2: what separated retrank from similarly accurate competitors was knowing when it had answered well. The cleanest evidence is a single comparison: team BITEM reached Accuracy 0.92 — close to our 0.95 — but HMR 0.49, roughly half of ours. Two systems of comparable answer quality, a 2× gap in HMR: on this comparison the difference lies in confidence calibration, not answer quality. We show that most competitors fell into one of two calibration failure modes, and that retrank is the team whose average confidence tracked its accuracy most tightly, and did so at the highest accuracy.

The LLM-judging question, up front. All R2C2 relevance labels are LLM-generated (official qrels from Qwen-3.5 and GPT-5.5), and, like every team, we use LLMs — the task permits and presumes it. So the question is not whether using an LLM was fair (it was universal) but whether an LLM-*judged* evaluation is trustworthy and whether our margin is method or model. We treat these as first-class research questions (Section 5), answered with cross-family agreement, out-of-sample tuning, and model-tier ablations.

Contributions.

(1) A calibration-first AC pipeline that sets confidence from an *independent verifier* rather than the generator, and the empirical finding that this — more than answer quality — is where HMR is largely won or lost (§3, §4.1).
(2) The first-place official result on all three metrics, with the diagnostic that most of the field splits into two calibration failure modes (§4).
(3) Evidence that the win is *method*, not *pool*: we won on a competitor's passages while our own retrieval ranked last

of 22 on PR, and phase-1 (retrieval) standing did not transfer to phase-2 (AC) across teams (§4.2).

(4) A dedicated robustness analysis against LLM-based evaluation: held-out cross-family self-evaluator validation, a controlled *calibration-transfer* experiment in which our confidence layer raises *competitors'* HMR *under our self-evaluator* (mean +0.065 across four teams, slr2c2 quarantined), a *base-model swap* reproducing the calibration gain when the streamlined pipeline runs on three GPT models *under a fixed GPT judge* (3/3 positive, one significant; mean ΔHMR+0.27), and a no-LLM encoder-only lower bound (§5, §5.3).

2 Task and Metric

We keep this brief; the task overview [1] is authoritative. For each of 65 topics a system returns an answer, a confidence $c \in [0, 100]$, and nuggets each citing a passage from a pooled PR run. One topic is dropped from official scoring (an organiser-side judgment gap), so all *official* metrics are computed over 64 topics; our self-evaluated surrogate experiments (self-eval, calibration-transfer, GPT-swap, no-LLM) score all 65, and we note n at each use. **Accuracy** is the fraction of correct answers; **MNP** is the mean precision of cited nuggets; **HMR** [2] is the harmonic mean of an overconfidence reward R_{O} (high when the system is not confidently wrong) and an underconfidence reward R_{U} (high when the system is not needlessly diffident on correct answers). Because HMR is a harmonic mean, a system must score well on *both* rewards; collapsing either one collapses HMR. Crucially, all relevance labels are *model-generated* (official qrels from Qwen-3.5 and GPT-5.5); we return to the consequences in Section 5.

3 A Calibration-First Pipeline

Our system is answer-first with verification: a single candidate answer is proposed early, then used to focus nugget extraction and, crucially, confidence estimation. We describe the stages operationally so the calibration layer can be reproduced.

Stage 0 — Pool. Aggregate top-K passages from selected teams' PR runs, deduplicate by `(doc_id, text-prefix)`, rank by a cross-encoder [16], cap at 50.

Stage 1 — Candidate answer. The generator reads the pool and returns (`answer`, c_{self}), instructed to answer only from the passages.

Stage 1.5 — Answer-conditional refinement. Re-rank the pool conditioned on the candidate (cross-encoder on ⟨question | answer⟩, or pointwise "supports-the-answer" scoring); keep top 30.

Stage 2 — Targeted nuggets. Emit 3–8 atomic claims supporting the candidate, each citing one passage.

Stage 3 — Bogus filter. Entailment-check every (passage, nugget) pair; drop unsupported nuggets (this drives MNP).

Stage 4 — Independent verifier. A separate call derives an answer using *only* the entailed nuggets (no passages) and compares it to the candidate, yielding `match_score`$\in \{0, 1, 2\}$.

Table 1: Best run per team, official metrics (7 teams; higher is better). retrank is first on all three and sweeps the top four on HMR and MNP.

Team	Best HMR	Best Acc	Best MNP
retrank	**0.974**	**0.969**	**0.840**
WaterlooClarke	0.761	0.938	0.686
ORG (organisers)	0.760	0.734	0.538
slr2c2	0.729	0.547	0.173
Error404	0.518	0.952	0.470
BITEM	0.492	0.922	0.781
hit-u	0.184	0.813	0.477

Table 2: retrank's four official runs. The calibration layer keeps R_{U} high across all four; HMR tracks R_{O}, and even the weakest (AC-4) exceeds the field's best (0.761).

Run	role	R_{O}	R_{U}	Acc	HMR
AC-1	saturation	0.95	1.00	0.953	**0.974**
AC-2	main system	0.95	0.976	0.953	0.963
AC-3	accuracy hedge	0.85	0.964	**0.969**	0.903
AC-4	6-team pool	0.79	0.962	0.891	0.868

Stage 5 — Confidence & refusal. Final confidence combines c_{self}, `match_score`, and verifier agreement; the system refuses on weak signals. The key design choice is that confidence is anchored to the *independent verifier*, not the generator's self-report.

Stages 3–5 are the *calibration layer*: a wrapper that can, in principle, sit on any answer engine (we test this cross-system in §5.1). Our central methodological claim is that this layer appears to contribute more than the base model on HMR.

Tuning on out-of-sample synthetic topics. To avoid fitting design choices to the 65 official topics, we selected the pipeline's free parameters — most importantly the verifier-score abstention threshold — on **val250**, a held-out set of 248 synthetic topics sampled from $\sim 15{,}000$ we generated across 50 challenge types (e.g. multi-hop, negation, vocabulary-gap, real-vs-fictional) over 559 movies. The official topics were never used for threshold or configuration selection. This is a deliberate generalisation guard: the thresholds that produced our submitted runs were chosen to maximise HMR on topics the official set never saw.

4 Results: Official Standing

Table 1 gives the best run per team on each official metric. retrank is first on all three.

Our four runs are deliberately diverse bets rather than near-duplicates (Table 2): a confidence-saturation recalibration (AC-1), the clean main system (AC-2), an accuracy-favouring variant (AC-3, highest accuracy of any run at 0.969), and a judge-divergent six-team-pool hedge (AC-4). All four share the calibration layer, so R_{U} stays high (≥ 0.96) while R_{O} — and hence HMR — varies with the design.

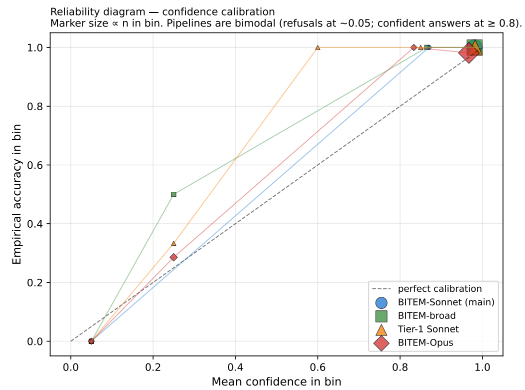

Figure 1: Reliability diagram for retrank's submitted runs (main system AC-2 highlighted): reported confidence vs. empirical accuracy per bin. Confidence tracks accuracy (near-diagonal), not merely in the mean.

4.1 Calibration, not correctness, is the lever

Most of the field splits cleanly into two calibration failure modes (Table 3). *Overconfident* systems (ORG, slr2c2) answer at ~0.9 confidence but are only ~0.53 accurate, so confident-wrong answers collapse R_{O} (ORG's best-accuracy-style run: confidence 0.90, accuracy 0.53, $R_{\mathrm{O}} = 0.09$). The lone *sandbagger* (hit-u) reports near-zero confidence everywhere — average confidence 0.016 — and although it is 0.81 accurate, this collapses R_{U} to 0.02 and HMR to 0.036. retrank is the team whose *average confidence tracks its accuracy most tightly* (best run AC-1: confidence 0.956 vs. accuracy 0.953); the one other well-calibrated team, WaterlooClarke, tracks less tightly and at lower HMR. Because a near-zero *mean* calibration error (+0.003) can coexist with poor per-level reliability, we also report a reliability diagram (Fig. 1) [8]: confidence tracks accuracy across bins, keeping R_{O} and R_{U} simultaneously high.

We stress that we did *not* win by being modest [6]. Our winning runs answer ~95% of topics correctly at high confidence; the always-refuse strategy that also scores high HMR does so at near-zero accuracy, whereas we reach top HMR *with* the field's best accuracy. The 2× HMR gap over BITEM at comparable accuracy (Table 1) supports calibration as the main differentiator among high-accuracy systems.

The R_{O} bottleneck. Why does calibration dominate? Across our 16 development configurations R_{U} occupies a narrow high band ($[0.79, 0.98]$) while R_{O} ranges widely ($[0.45, 0.95]$), so at this operating point HMR is far more sensitive to R_{O}. The official scores confirm it: across our four submitted runs $R_{\mathrm{U}} \in [0.96, 1.00]$ but $R_{\mathrm{O}} \in [0.79, 0.95]$, and HMR tracks R_{O}. Every winning design choice we identify — verifier-anchored confidence, answer-conditional refinement, refusal on weak evidence — primarily *raises* R_{O} (suppresses high confidence on wrong answers) rather than lifting R_{U}. The practical lesson for confidence-scored RAG is therefore asymmetric: *engineer to suppress confident errors, not to make correct answers more confident.*

Table 3: The two failure modes. R_{O} collapses under overconfidence (low accuracy at high confidence); R_{U} collapses under sandbagging (near-zero confidence). retrank keeps both high. (Representative official runs.)

Run	mode	conf.	acc.	R_{O}	R_{U}
retrank-AC-1	calibrated	0.956	0.953	0.95	1.00
ORG-AC-1	overconfident	0.905	0.531	0.09	0.90
hit-u-AC-4	sandbagging	0.016	0.813	0.99	0.02

Table 4: Phase-1 (PR) and phase-2 (AC) standing are strongly anti-correlated across the six teams with both runs (Spearman $\rho_s \approx -0.94$): the best retriever answers worst, and retrank (last on PR) wins AC.

Team	best PR nDCG@20 (rank)	best AC HMR (rank)
hit-u	0.438 (1)	0.184 (last)
BITEM	0.422 (2)	0.492
Error404	0.346 (3)	0.518
WaterlooClarke	0.318 (4)	0.761
ORG	0.309 (5)	0.760
retrank	**0.114 (last)**	**0.974 (1)**

What overconfidence looks like. BITEM's best-HMR run (BITEM-AC-2, the run in our core comparison) makes the failure concrete. Its confidence distribution is neither flat nor trivial — confidences span 0–100 with mean 83.5 — but it is systematically *high*: the median is 97 and 46% of answers carry confidence 100. Near-maximal confidence on the ~8% of topics it answers wrong is enough to collapse R_{O} to 0.35, halving HMR despite 0.92 accuracy. The calibration failure is genuine, systematic overconfidence, not an artefact of a degenerate strategy — which is exactly what the thesis predicts.

4.2 The win is the pipeline, not the pool

Our winning runs use *BITEM's* passages, not our own. Our own PR runs ranked *last of 22* on nDCG@20 under the AC-derived relevance labels (0.09–0.11 vs. 0.44 for the top run). More strikingly, phase-1 and phase-2 standing are strongly *anti*-correlated across the six teams with both a PR and an AC run (Spearman $\rho_s \approx -0.94$; slr2c2 is omitted for lack of a comparable PR run): the best retriever (hit-u, PR rank 1) is the worst answerer (AC rank last), and the worst retriever (retrank) wins AC. We read this with care: because PR is re-scored by how often AC systems cited each passage, and our AC preferentially cited BITEM's passages, the inversion is partly a *mechanical* consequence of citation choices rather than an independent measure of retrieval quality — we therefore do not claim retrieval was useless. The defensible, weaker conclusion is that *phase-1 standing did not transfer to phase-2*: our advantage rode on the pipeline operating over a competitor's pool, not on our own retrieval. Consistently, adding weaker passages to the strong BITEM pool (a ten-run multi-team pool) *lowered* HMR from 0.963 to 0.868: dilution, not enrichment.

4.3 Internal ablations

Beyond the cross-team comparison, a development sweep over 16 base configurations (val250-tuned) isolates which design choices move HMR, and confirms they act through R_O. We grade each by whether its 95% bootstrap CI on ΔHMR excludes zero (percentile intervals from B=5000 topic-level resamples, recomputing R_O, R_U, and HMR within each resample, with contrasts paired on topics). Because these contrasts are drawn from an exploratory sweep, the intervals are descriptive and *not* multiplicity-adjusted; we label the evidence grade accordingly.

Robust. Per-stage model tier matters most at the candidate-answer stage (ΔHMR=−0.13 Sonnet→Haiku, CI excludes zero) and least at the verifier (−0.05); the all-Sonnet→all-Haiku regression (−0.17) and the naive-pooling regression are also robust.

Suggestive. Answer-conditional pool refinement helps (+0.085, CI [−0.045, +0.44]); a strong single-team pool beats naive multi-team pooling (−0.11, CI [−0.46, +0.04]) — an effect of pool *quality*, not count.

Preliminary. Answer-first beats nugget-first on a 20-topic head-to-head (equal accuracy, −0.29 HMR for nugget-first, *entirely* attributable to R_O).

Verifier-based abstention is context-dependent: on a noisy multi-team pool it adds +0.04 HMR over never abstaining, but on the clean BITEM pool it adds nothing. The win on our submitted (clean-pool) runs therefore comes from confidence *calibration*, not abstention per se — and every one of these levers moves R_O, corroborating the bottleneck of §4.1.

5 Robustness to LLM-Based Evaluation

Every R2C2 team used large language models: the task is LLM-based RAG and the rules place no restriction on model choice, so using an LLM — Claude included — is not an advantage anyone can contest. The legitimate questions are therefore not about *permission* but about *validity* and *attribution*: (i) whether an evaluation whose ground truth is itself LLM-generated (Qwen-3.5, GPT-5.5) yields trustworthy scores, and (ii) whether our margin reflects our method or merely raw model strength. We address both.

Objection A: shared-judge bias. "You optimised to an LLM, and the judges are LLMs." We tuned against a Claude self-evaluator; the official labels come from *Qwen-3.5 and GPT-5.5*. These are different model families, so any agreement is not self-agreement. Our Claude self-evaluator predicted the *official* HMR of our two main runs to within **0.0004** (self 0.974/0.963 vs. official 0.9744/0.9626). Cross-family agreement at this precision is evidence that the calibration we optimised is real signal, not an artefact of a judge scoring its own family. We report the same self-vs-official comparison for our other runs, where agreement is looser (≤ 0.02), and we are explicit (§7) that the two tightest points are runs we tuned against, so this is corroborating rather than held-out evidence.

A held-out test: predicting competitors' HMR.. To break the circularity, we scored five competitor runs we did *not* tune against — spanning the official HMR range — with a lightweight version of our Claude self-evaluator and compared its predictions to the official Qwen/GPT scores (Table 5). Agreement is *weak*: Spearman ρ=0.60 but on only n=5 runs, so it is not significant (exact permutation $p\approx0.35$); mean absolute error is 0.12, with three of five runs predicted within 0.01–0.13 and one large miss (slr2c2, whose well-calibrated low-confidence strategy our simplified judge under-scored). The self-evaluator therefore shows a non-circular but statistically weak cross-family *signal* — it orders the extremes correctly — rather than demonstrated validity, and it is *not* tightly calibrated on other teams' runs. We treat it as a small diagnostic: the near-perfect agreement on our own runs above partly reflects tuning, whereas the held-out agreement is positive in this sample but not statistically established.

Table 5: Held-out cross-family check: our (lightweight) self-evaluator's predicted HMR vs. the official Qwen/GPT HMR on five competitor runs we did not tune against (weak, n=5: ρ=0.60, permutation $p\approx0.35$, MAE 0.12).

Competitor run	our self-eval	official
WaterlooClarke-AC-2	0.750	0.761
hit-u-AC-1	0.187	0.184
Error404-AC-3	0.504	0.518
BITEM-AC-2	0.621	0.492
slr2c2-AC-4	0.286	0.729

Within-family judge stability. We additionally re-judged all four submitted runs with a second Claude judge (Opus 4.7). Run ordering is preserved and the two main runs move by ≤ 0.001 HMR; only the hedged runs move materially (AC-3 +0.03, AC-4 +0.06 Sonnet→Opus), and the official scores fall between the two judges' estimates — judge disagreement is concentrated on the hedged runs, not the main system.

Objection B: you just used a stronger model. If raw model capability drove the win, scaling the model up should help. It does not: with prompts held fixed (developed on Sonnet), an *all-Opus* pipeline *regressed* HMR by 0.18 and an *all-Haiku* pipeline by 0.17 (self-eval; bootstrap CIs exclude zero). We read these cautiously — because prompts are not re-tuned per tier, they measure *naive model substitution*, not intrinsic capability (§7), so we do *not* claim a larger model is worse. Decomposing the Opus regression makes the mechanism precise: it is concentrated in the overconfidence reward (ΔR_O=−0.27), *not* accuracy (ΔAcc=−0.05). Opus answers nearly as accurately, but under the Sonnet-tuned verifier prompt its confidence fails to separate right from wrong answers — and this does not recover under post-hoc recalibration (HMR 0.79 → 0.78), so it is a confidence-*discrimination* effect localised to the verifier, not inferior answering. The claim we rely on is one-directional and deliberately weak: subtracting the all-Haiku penalty from our main system still leaves HMR ≈ 0.79, near the best non-retrank run (0.761) — **our actual pipeline, run on the cheapest Claude tier, would stay in the competitive range of the field-leading team**. The ∼ 0.03 margin is far smaller than our self-evaluator's cross-family error on held-out competitor runs (MAE 0.12, §5), so we do *not* claim even parity — only that a far cheaper model keeps the pipeline near the top of the field, so the margin is not bought by model strength.

Objection C: you overfit to the test set. We selected the pipeline's parameters (notably the abstention threshold) on **val250**, 248 held-out synthetic topics, and never on the 65 official topics (§3). The official set was thus held out from configuration selection, and the tight self-vs-official agreement above is measured on topics not used to choose thresholds. This does not escape the LLM loop — the synthetic topics are themselves Claude-generated — but it does rule out fitting to the specific test topics or their specific official-judge outputs.

Why this is a calibration layer, not a model. Objection B's ablations show the gain is not bought by raw model strength; Section 4.1 shows it is in the confidence layer; and BITEM demonstrates that a comparable answer engine without that layer scores half our HMR. The natural reading is that the lever is a *model-agnostic calibration wrapper*. We test this directly.

We test this two ways: a controlled calibration-transfer experiment (below) and a no-LLM lower bound (§5.3).

5.1 Calibration transfer across systems

The no-LLM committee swaps both the answerer and the calibrator's form; a cleaner test of whether *our specific confidence layer* is the moat is to hold a competitor's answers fixed and replace only their confidence with ours. For five competitor runs we ran our verifier (Stage 4) on their submitted answers and applied our Stage-5 rule, then re-scored HMR under our self-evaluator (Table 6; the official judge cannot be re-run). We *quarantine* slr2c2: although it shows the largest gain (+0.38), it is exactly the run our self-evaluator mis-scores worst (Table 5: self 0.29 vs. official 0.73), so a gain measured inside that unreliable regime cannot be trusted and we exclude it from the summary. Across the other four teams our layer improves HMR for three and the mean gain is +0.065, concentrated on the *overconfident* systems (Error404 +0.14, BITEM +0.13) with essentially no change for the already-calibrated WaterlooClarke (+0.01). The lone regression is the sandbagger hit-u (−0.02): our layer only *suppresses* overconfidence, so it cannot help a system that is already under-confident — exactly the asymmetry the R_{O} bottleneck predicts. Baselines differ slightly from Table 5 because the two lightweight evaluators use different correctness prompts, but the within-experiment original-vs-transfer comparison holds the evaluator fixed, so it is unaffected by that baseline offset — though, like every delta here, it remains *conditional on the surrogate's correctness labels*: a fixed judge guarantees ON and OFF share labels, but shared Claude-family agreement between our verifier and self-evaluator could still flatter the layer. Because the answers come from other teams' systems, this is controlled evidence that the confidence wrapper transfers across systems *under our self-evaluator*; genuine cross-base-model-family transfer within our own pipeline remains unproven.

Table 6: Calibration transfer: holding each competitor's *answers* fixed and replacing only their confidence with our verifier-capped layer. Our layer raises HMR for the overconfident teams (via R_{O}) and cannot help the sandbagger (hit-u). slr2c2 is quarantined (our self-evaluator mis-scores it; see text). Scored under our self-evaluator, not the official judge; ΔHMR is computed from unrounded values.

Team run	Acc	HMR orig	HMR +layer	ΔHMR
slr2c2-AC-4	0.68	0.243	0.619	**+0.376**
Error404-AC-3	0.71	0.553	0.697	+0.144
BITEM-AC-2	0.92	0.599	0.725	+0.125
WaterlooClarke-AC-2	0.83	0.820	0.827	+0.007
hit-u-AC-1 (sandbag)	0.81	0.197	0.179	−0.018

Table 7: Base-model swap: streamlined pipeline run on three OpenAI models (fixed GPT-5.5 judge). Calibration raises HMR for all three, always via R_{O}; ΔHMR with 95% paired-bootstrap CI (* excludes zero). Direction consistent (3/3) but only GPT-4.1-mini is individually significant — the accurate GPT-5.6 variants have too few errors to power the test. ΔHMR is computed from unrounded values. Model ids as exposed on the account.

Answer/verifier model	Acc	HMR off	HMR on	ΔHMR [95% CI]
gpt-5.6-sol	0.939	0.444	0.599	+0.156 [−0.03,+0.63]
gpt-5.6-luna	0.954	0.577	0.717	+0.140 [−0.04,+0.49]
gpt-4.1-mini	0.923	0.399	0.899	+0.500 [+0.10,+0.81]*
mean	—	—	—	+0.265

5.2 Base-model swap: the pipeline on GPT

The transfer test above keeps each competitor's answers fixed and swaps only the confidence layer; the stronger test is to run the pipeline *end-to-end* with a *non-Claude* answer/verifier model. We reimplemented a *streamlined* version (candidate → nuggets → verifier → Stage-5, using the top-20 pool passages by rank *without* the submitted system's cross-encoder re-rank, 50-passage cap, or Stage-3 entailment filter) with three OpenAI models on the 65 topics and BITEM pool, scoring correctness with a fixed GPT-5.5 judge (Table 7). The ON and OFF runs share identical answers, nuggets, and correctness labels (including refusals) and differ *only* in confidence; holding the judge fixed keeps those labels constant across ON/OFF, so the paired HMR delta isolates the calibration rule. All three models improve, always via R_{O} — the same signature as Claude. The direction is consistent (3/3 positive) but the per-model effects are *underpowered*: because the two GPT-5.6 variants are ≥ 93% accurate they have only 3–4 incorrect answers to recalibrate, so their R_{O} gains are positive with 95% bootstrap CIs that cross zero; only GPT-4.1-mini (more errors, larger effect) is individually significant. Further caveats: correctness is scored by a GPT judge, not the official Qwen+GPT ensemble, so absolute HMR here is context, not a like-for-like ranking (only GPT-4.1-mini-ON reaches the field's 0.761; none reaches our Claude 0.963). The claim we draw is narrow and supported: the verifier-cap rule improves HMR across three GPT configurations under a fixed surrogate judge — cross-family evidence (from Claude) that the calibration mechanism is not a Claude artefact. As with the Claude self-evaluator, a GPT judge scoring GPT answerers may share family-level agreement, so this is cross-family from Claude but not judge-independent.

5.3 A No-LLM Lower Bound

We rebuilt the pipeline with **no generative model**: a cross-encoder reranker [16] over the winning pool, a *committee* of extractive QA encoders (RoBERTa [18], ELECTRA [19], MiniLM [20], BERT, TinyRoBERTa, all SQuAD2 [17]) that vote span-by-span, and a confidence calibrator over the answerability margin and committee agreement, fit on the held-out synthetic val250 set. This calibrate-then-abstain design follows Kamath et al. [7], who train a classifier to predict QA errors; we extend it to the AC/HMR setting.

Voting scales accuracy, then plateaus. Committee voting — a discrete analogue of self-consistency [22] — raises accuracy from 0.42 (one model) to 0.52 (three), then saturates: a fourth and fifth answerer add nothing (Table 9), so the extractive ceiling is ≈ 0.52. Per-answer calibration degrades as the committee grows (margin AUC 0.85 at two answerers to 0.74 at five; Table 9), so HMR peaks at a two-model committee. At answer-everything the committee scores HMR 0.66 *under our self-evaluator*. Because this is a surrogate score, not an official run, we do not insert it into the official ranking; but scored by the *same* self-evaluator, the committee (HMR 0.66, Acc 0.52) draws level with the far more accurate BITEM (HMR 0.62 under the same judge, Table 5; Acc 0.92, Table 6). This is the calibration thesis in miniature: a *calibrated but inaccurate* system matches an *accurate but overconfident* one, under a fixed judge, precisely because HMR rewards the alignment of confidence with correctness, not correctness itself.

Which non-LLM techniques help? Table 10 sweeps the techniques available without a generative model. *Confidence is aleatoric and cheap*: the answerability margin predicts correctness at AUC up to 0.85 (two-model committee), and adding second-order feature interactions — in the spirit of SmoothHess [12] and integrated-Hessian [13] interaction analysis — lifts the larger (three-model) committee's margin AUC from 0.766 to 0.803 (Table 10), but only under strict feature parsimony. *Every variance-based uncertainty estimator failed*: MC-dropout [9] (AUC 0.53), deep-ensemble score dispersion [10] (AUC 0.51); a last-layer Laplace approximation [11] targets the identical posterior-predictive variance and so we predict it behaves the same, though we did not run it. The useful signal is the point estimate, not its spread. *Abstention should be coverage-controlled* (conformal [14, 15]), not HMR-maximised — the latter games the metric to answering 17% of topics at 0.14 accuracy. *Accuracy needs scale*: a small generative reader (flan-t5-base [24]) reaches only 0.48, *below* the extractive committee, over-generalising where extraction stays precise.

So the LLM's measured contribution is answer accuracy at scale — not the confidence layer, and not, in these experiments, alignment with an LLM judge. "You only won because you used a strong LLM" reduces to "the LLM answers harder questions," which is the task.

Statistical power. Following Sakai's advocacy for reporting significance and power rather than bare point estimates [3, 4], we apply the randomised Tukey HSD test [3] (B=5000) to the committee ladder. The voting gain from one to three answerers is *suggestive but not significant* (ΔAcc=+0.108, p=0.14): voting fixes 12 wrong single-model answers but breaks 5 correct ones (net +7/65). A simulation-based power analysis shows the minimum accuracy difference detectable at n=65 with 80% power is ≈ 0.15 — above the observed gain — so this collection lacks the power to detect an effect of the observed size; the result is inconclusive, not a demonstrated null. We therefore report it as a trend; only the $0.52 \rightarrow 0.94$ committee-vs-LLM accuracy gap (both self-evaluated on n=65) far exceeds the detectable threshold, so it is robust to the judge and to the 64-vs-65-topic denominator.

Table 8: Per-type accuracy (both systems self-evaluated on the 65 typed topics), committee-3 vs. LLM (rule-based typing). The LLM leads on every type; its margin is largest on count/aggregation questions.

Question type	n	committee	LLM
factoid	27	0.59	1.00
person	19	0.58	0.95
count ("how many")	14	**0.29**	0.86
title	5	0.60	0.80

Table 9: No-LLM committee ladder (answer-everything). Voting scales accuracy to a ≈ 0.52 plateau; HMR peaks at a two-model committee as calibration degrades. Committee rows are self-evaluated (n=65); the LLM pipeline row is official (n=64; our self-eval agrees with official to 0.0004, §5), shown for reference.

System	Accuracy	HMR	margin AUC
Committee-1 (RoBERTa)	0.415	0.629	0.849
Committee-2 (+ELECTRA)	0.462	**0.677**	0.85
Committee-3 (+MiniLM)	**0.523**	0.659	0.766
Committee-5 (+BERT,+TinyRoBERTa)	0.523	0.644	0.736
LLM pipeline (retrank-AC-2)	0.953	0.963	—

Per-topic failure analysis. Following the organisers' request for per-topic analysis, we heuristically type the 65 questions (rule-based, no LLM) and compare committee and LLM accuracy by type (Table 8). The LLM dominates *every* type, but its advantage is largest exactly where reasoning is required: on *count* questions ("how many…"), which need an aggregation no extracted span can express, the committee collapses to 0.29 against the LLM's 0.86. Even on plain factoids the committee reaches only 0.59 (LLM 1.00), because the answer is not always a clean extractable span. No question type is competitive for extraction; the generative advantage is pervasive and peaks on aggregation — a concrete localisation of the reasoning gap.

6 What Did Not Work

Four interventions we expected to help, hurt. *(The model-tier ablations below are self-evaluated and hold prompts fixed at their Sonnet-developed form, so they measure naive substitution under our judge, not intrinsic model capability — see §7.)* (1) *Scaling up.* An all-Opus pipeline regressed HMR by 0.176. Decomposition is instructive: the drop is almost entirely in the overconfidence reward (ΔR_O=−0.27), not accuracy (ΔAcc=−0.05) — Opus answers nearly as well but, under the Sonnet-tuned verifier prompt, reports high confidence

Table 10: Non-LLM technique sweep (self-evaluated surrogate, n=65). Confidence is recovered cheaply; variance-based uncertainty fails; accuracy needs scale.

Technique	Metric	Result
Committee voting (1→3)	Accuracy	0.42→0.52 (plateau)
Interaction calibrator (2nd-order)	disc. AUC	0.766→**0.803**
Discrete agreement	AUC	0.76
Ensemble score-variance	AUC	0.51
MC-dropout variance	AUC	0.53
Last-layer Laplace	*untested*	pred. ≈0.5
Conformal abstention	—	coverage knob
Small generative reader (flan-t5)	Accuracy	0.48 (<0.52)

on its errors, and this does not recover under post-hoc recalibration (0.79 → 0.78). The regression is confidence *discrimination* at the verifier, not answer quality. (2) *Scaling down.* All-Haiku regressed by 0.170; the useful configuration is *selective* tiering — a strong model at the candidate-answer stage (the most model-sensitive, ΔHMR = −0.13 Sonnet→Haiku) and cheaper models downstream (the verifier is least sensitive, −0.05) — not uniform scale in either direction. (3) *Naive multi-team pooling.* Enlarging the strong single-team pool with weaker passages lowered HMR (0.963 → 0.868); more evidence was dilution, not enrichment. (4) *Per-topic oracle pool selection.* Picking, per topic, the pool whose verifier most strongly agreed with its candidate — an apparently free "oracle" — also reduced HMR: verifier agreement is a good *ranking* signal but a biased *selection* criterion, favouring confidently-wrong candidates on hard topics.

7 Limitations

We are deliberately conservative about what the result does and does not show.

- **LLM-judged ground truth.** Accuracy, MNP, and HMR all rest on model-generated labels (Qwen-3.5, GPT-5.5). Our cross-family agreement (§5) mitigates but does not eliminate the concern; none of these numbers are human-validated.
- **Self-evaluator validation is partly in-sample.** The tightest self-vs-official agreement is on runs we tuned against the self-evaluator. A fully non-circular test is to predict *other teams'* official HMR from our self-evaluator; we report this where data permit and flag it as the cleaner validation.
- **Synthetic tuning data is also LLM-generated.** val250 mitigates overfitting to the 65 official topics, but the synthetic topics were produced by Claude (Sonnet generator, Haiku in-loop judge), so tuning generalisation is demonstrated within a Claude-imagined distribution, not against human-authored topics.
- **Cross-family transfer is now shown, under a surrogate judge.** The calibration layer's model-agnosticism is demonstrated two ways: the calibration-transfer experiment (§5.1) lifts three of four non-quarantined *other teams'* runs, and the base-model swap (§5.2) reproduces the calibration gain when the *streamlined* pipeline runs end-to-end on three GPT models (3/3 positive ΔHMR). The residual caveat is the judge: both are scored by our own self-evaluator or a GPT judge, not the official Qwen+GPT ensemble, which we cannot re-run. A non-Claude pipeline scored by the official judge is the only piece still outside our reach.
- **Model-tier ablations hold prompts fixed.** The all-Opus and all-Haiku configurations reuse the prompts developed on Sonnet, changing only the model. The resulting regressions therefore measure *naive model substitution without re-tuning*, not intrinsic model capability; we do not claim Opus is intrinsically worse, only that the pipeline is co-adapted to its model and cannot be re-tiered for free. The complementary claim — that our (Sonnet-prompted) pipeline run *on Haiku* would remain *competitive with the field-leading team* (in the field's competitive range, not a decisive lead) — is unaffected, since it evaluates the actual deployed pipeline on a cheaper model.
- **Small sample and evidence grades.** n = 64 topics. Some findings are suggestive (answer-conditional refinement, CIs cross zero) or preliminary (answer-first vs. nugget-first, 20 topics); we mark them as such rather than headlining them.

8 Reproducibility

We release code and data; the operational specifics are: **Pool.** BITEM-PG-1 + BITEM-PG-2, deduplicated by `(doc_id, 64-char prefix)`, re-ranked by `cross-encoder/ms-marco-MiniLM-L-12-v2`, capped at 50 passages. **Models.** Claude Sonnet 4.6 at all stages of the submitted runs; Haiku 4.5 / Opus 4.7 for the tier ablations; the self-evaluator is Sonnet 4.6 with Opus 4.7 cross-validation. **Verifier (Stage 4).** A separate call re-derives an answer from the *entailed nuggets only* and compares it to the candidate, emitting `match_score`$\in \{0, 1, 2\}$ (0 disagree/refuse, 1 partial, 2 full). **Confidence (Stage 5), the exact rule.** If there are no entailed nuggets or the answer is empty, refuse (`conf=` 5); else cap the model's self-confidence at 25 if `match_score=` 0, at 60 if = 1, and leave it unchanged if = 2. The abstention behaviour and configuration were selected on val250; the 65 official topics were never used for tuning.

9 Related Work

Confidence-scored answering and modesty. HMR and the "modesty" framing of over- and under-confidence are due to Sakai and the R2C2 organisers [2, 5, 6]; unlike aggregate calibration measures such as expected calibration error [8], HMR penalises the two error directions separately. Our paper is a case study in optimising it. **Selective prediction and abstention.** Deciding *when* to answer has a long history [7]: Kamath et al. train a classifier over model features to predict errors and abstain — the design our no-LLM calibrator follows. Conformal prediction [14, 15] gives distribution-free coverage control, which we explore as a principled alternative for setting abstention thresholds; the submitted runs used HMR-maximised thresholds selected on val250. **Calibration and uncertainty.** Post-hoc calibration [8] and Bayesian uncertainty — MC-dropout [9], deep ensembles [10], and Laplace approximations [11] — are standard tools; we find that variance-based estimators we tried (MC-dropout, deep ensembles) add nothing on this task and the point-estimate margin dominates. Second-order feature-interaction analysis [12, 13] inspired our interaction-aware calibrator. Our retrieval and answering components build on cross-encoder re-ranking [16] and extractive QA [17–20]. **RAG evaluation and LLM judges.** R2C2 pools passages across teams for retrieval-augmented answering [21]. Because its relevance labels and answer judgments are produced by LLMs [23], we treat judge validity as a first-class concern, and our committee voting is a discrete analogue of self-consistency [22].

10 Conclusion

R2C2's Answering-with-Confidence subtask is a calibration problem wearing a QA costume. The teams whose confidence did not track their accuracy fell into one of two failure modes and lost, sometimes despite excellent retrieval or competitive accuracy. retrank treated it as calibration from the start: an answer-first pipeline whose confidence comes from an independent verifier, winning first place on all three official metrics, with all four of its runs above every non-retrank run on HMR. Because the win was scored by LLMs, we tested it against that fact rather than around it, and found the margin is not explained solely by using the largest Claude tier. Under our surrogate judges, the calibration layer itself transfers: it lifts three of four non-quarantined competitors' runs (self-evaluator) and, run on three GPT models (fixed GPT judge), reproduces the calibration gain (3/3 positive, one significant). What remains — confirmation under the *official* Qwen+GPT judge, and the residual limitations of §7 (64 scored topics, LLM-generated labels, Claude-tuned thresholds) — separates a strong participant result from a fully general method claim.

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